DNS Tunneling Detection — Project Report

Author: _your name_ **Course**: _course name_ **Date**: _submission date_ **Repository**: <https://github.com/df-DNS-Tunneling-Detection/DNS_Tunneling_Detection>

Abstract

This project builds a machine-learning pipeline that detects DNS tunneling traffic — a covert-channel technique attackers use to exfiltrate data or run command-and-control over the DNS protocol. We train and compare three detectors on the CIRA-CIC-DoHBrw-2020 dataset: two tree-based classifiers (Random Forest, XGBoost) and a deep-learning baseline (a Multi-Layer Perceptron). On a balanced held-out test split of 998 flows, XGBoost reaches **0.999 accuracy / F1 and 1.000 ROC-AUC**, Random Forest reaches **0.998 / 1.000**, and the MLP reaches **0.992 / 0.999**. The repository ships pretrained `.pkl` models, a bundled 10 000-row sample of the dataset, an inference CLI, and self-contained Colab notebooks so anyone can reproduce the results in minutes without registering for the source dataset.

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1. Introduction

1.1 Topic and Problem

The **Domain Name System (DNS)** is one of the oldest and most permissive protocols on the internet. Almost every firewall allows outbound traffic to port 53 (and now port 443 for DNS-over-HTTPS), and very few networks deeply inspect DNS payloads. Attackers exploit this trust by **encoding arbitrary data into the subdomain portion of DNS queries**, turning the protocol into a bidirectional covert channel between a compromised host and an attacker-controlled name server. This technique is called **DNS tunneling**, and it underpins production attack tools such as `iodine`, `dnscat2`, and `dns2tcp`.

Two common attacker goals are served by DNS tunneling:

- **Command-and-control (C2)** — malware receives instructions disguised as DNS responses.
- **Data exfiltration** — sensitive data is chunked, base32-encoded, and sent as a stream of unique subdomain queries.

Signature-based defences (Suricata / Snort rules against known C2 domains) are brittle: any change to the encoding or destination defeats them. Volumetric thresholds are unreliable because slow tunnels intentionally stay below detection limits. With the rise of **DNS-over-HTTPS (DoH)**, queries are encrypted, so the resolver is the only vantage point with visibility into query content — making a learning-based detector that operates on flow metadata especially valuable.

1.2 Objectives

The project sets out to:

1. Build a reproducible ML pipeline that ingests real DNS-tunneling datasets and outputs trained binary classifiers.
2. Evaluate at least three model families — two tree-based ensembles and one neural network — under identical preprocessing and an identical train / validation / test split.
3. Quantify the contribution of each input feature so that the detector is interpretable, not just accurate.

4. Ship pretrained models, a CLI, and a Colab notebook so that results can be reproduced in minutes by a third party.

1.3 Why this matters

DNS tunneling is rarely caught by standard firewall rules and is heavily used in real intrusions. According to the Canadian Institute for Cybersecurity, encrypted DNS traffic (DoH) is growing rapidly across consumer browsers, which simultaneously protects user privacy and removes the most common detection vantage point. A statistical learning approach that can detect tunneling from coarse flow metadata — without needing to decrypt content — directly addresses this operational gap.

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2. Literature Review

This section reviews prior work organised into four threads: classic statistical detectors, deep-learning detectors over raw query text, flow-based detectors over DoH, and detectors built on the CIC datasets we use.

2.1 Classic statistical / metadata-based detection

Born and Gustafson (2010), *Detecting DNS tunnels using character frequency analysis*, were among the first to formalise the observation that tunneled queries have **distinctively flat character distributions** because they encode encrypted payloads. They proposed a *chi*^2-style test against a reference English frequency distribution. The approach is fast and explainable but is defeated by any encoder that shapes its output distribution.

Aiello, Mongelli and Papaleo (2013), *Basic classifiers for DNS tunneling detection*, evaluated NN, SVM, and RBF classifiers on synthetic and captured tunneling traffic, reporting F1-scores in the 0.95–0.99 range on small datasets. Their key methodological lesson is that feature engineering (query length, entropy, label statistics) carries more weight than classifier choice on this problem.

Aiello et al. (2014), *Profiling DNS tunneling attacks with PCA and mutual information*, extended this work with dimensionality reduction and showed that two or three engineered features can already produce highly separable classes — foreshadowing the near-perfect numbers seen on modern datasets.

2.2 Character-level deep learning

Yu, Pan, Hu, Nascimento and De Cock (2018), *Character-level based detection of DNS tunneling*, fine-tuned character-level convolutional and LSTM networks directly on query strings, removing the need for hand-crafted features. On their captured dataset they reported F1 around 0.99 and showed that the network learns features broadly equivalent to entropy and length. The drawback is the larger model size and a need for raw query text — unavailable when traffic is DoH-encrypted.

Liu et al. (2019), *A byte-level CNN for malicious-domain detection*, generalised the approach to domain reputation and reported similar gains over hand-crafted-feature baselines.

2.3 Flow-level detection over DoH

MontazeriShatoori, Davidson, Kaur and Lashkari (2020), *Detection of DoH tunnels using time-series classifiers*, are the authors of the **CIRA-CIC-DoHBrw-2020 dataset** we use. They proposed a two-layer pipeline: Layer 1 separates DoH from non-DoH traffic; Layer 2 separates benign DoH from tunneling DoH. With Random Forest and a small LSTM they reported **accuracy and F1 above 0.99** on Layer 2 — precisely the layer this project targets.

Hjelm (2019), *A new needle and haystack: detecting DNS over HTTPS usage*, motivates the same Layer-1 task but does not address tunneling specifically.

2.4 Surveys and operational context

Buczak and Guven (2016), *A survey of data mining and machine learning methods for cyber-security intrusion detection*, places DNS-tunneling detectors in the broader IDS landscape and notes that decision-tree ensembles consistently outperform single classifiers on tabular network features — the same finding we observe.

Mockapetris (1987, RFC 1035) and Hoffman & McManus (2018, RFC 8484, DoH) specify the protocols and define the 63-character per-label and 253-character per-name limits that we exploit as features.

2.5 Where this project sits

Compared to the literature above, our project (i) uses the *same* dataset as MontazeriShatoori et al. (2020), so the numbers are directly comparable; (ii) explicitly adds a deep-learning baseline (MLP) so the **tree-ensemble vs. neural network** comparison is reproduced inside one repository under identical preprocessing; and (iii) ships a self-contained reproducibility kit (pretrained `.pkl` files, a bundled sample, and Colab notebooks). The methodology is otherwise standard — the contribution is the engineering and the side-by-side comparison.

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3. Methodology

3.1 Tools

Tool / library	Version used	Role
Python	3.10+	base runtime
pandas / numpy	2.x / 1.24+	data manipulation
scikit-learn	1.3+	RandomForestClassifier, MLPClassifier, StandardScaler, metrics, StratifiedKFold
XGBoost	2.0+	XGBClassifier
PyTorch	2.x	custom MLP architecture in deep_learning/04_mlp_e2e.ipynb
matplotlib / seaborn	3.7+ / 0.12+	figures
joblib	1.3+	.pkl persistence
Jupyter / Google Colab	—	notebooks for EDA and demo
GitHub	—	source hosting and reproducibility kit

3.2 Data collection

We evaluated three potential datasets and converged on one:

Dataset	Outcome

CIC-Bell-DNS-2021	Rejected. Malicious-domain classification (benign vs. malware / phishing / spam), not tunneling. The loader is retained as a side-utility.
CIRA-CIC-DoHBrw-2020	Selected. Real DoH tunneling traffic.
Synthetic	Kept as offline fallback.

CIRA-CIC-DoHBrw-2020 is released by the Canadian Institute for Cybersecurity. It contains DNS-over-HTTPS traffic from:

- **Benign**: regular browsing with Chrome / Firefox against Cloudflare and Quad9 DoH resolvers.
- **Malicious**: tunneling traffic generated with `dns2tcp`, `DNSCat2`, and `Iodine`.

The dataset is organised in two layers:

- **Layer 1**: DoH vs non-DoH traffic (out of scope for this project).
- **Layer 2**: benign DoH vs malicious DoH = tunneling. **This is what we use.**

File	Rows	Description
l2-benign.csv	19 807	Benign DoH flows
l2-malicious.csv	249 836	Tunneling DoH flows
Total	269 643	

Each row is one network flow with 35 columns: source/destination IP and port, timestamp, duration, byte counts, and 21 packet-size / packet-time / response-time statistics plus a `Label` column. The full malicious file is 148 MB — larger than the 100 MB GitHub limit. We therefore sample a balanced **10 000-row stratified subset** (5 000 benign + 5 000 malicious, seed 42, ~4.5 MB) and commit it as `data/sample/doh_sample.csv`.

3.3 Preprocessing

`src/preprocess.py` performs:

- **Auto-detection of dataset layout** — looks for query / label columns by name; falls back to numeric features.
- **CIC-Bell-DNS-2021 adapter** — filename-encoded labels.
- **CIC-DoHBrw-2020 adapter** — uses the `Label` column directly.
- **Deduplication** — removes duplicate rows.
- **NaN imputation** — median-imputes the small number of missing `ResponseTime*` values (flows that never received a response).
- **Stratified 80 / 20 train / test split** with `random_state=42`. For the MLP track we use 80 / 10 / 10 train / val / test so the network has a validation set for early-stopping / scheduler decisions.
- **Standardisation** for the MLP — `StandardScaler` fit on the training split only, then applied to validation and test.

3.4 Feature engineering

`src/features.py` extracts **11 metadata features** from a query string when one is available:

Feature	Intuition
query_length	Tunneled queries carry payload → longer

subdomain_length	Payload lives in subdomains
entropy	Shannon entropy — encoded payloads ≈ uniform distribution
bigram_entropy	Higher-order randomness
digit_ratio	Base32 / base64 encodings are digit-heavy
uppercase_ratio	Mixed-case alphabets in encoded text
unique_char_count	Wider character set → more random
vowel_consonant_ratio	Natural domains have English-like vowel ratios
label_count	Many short labels can indicate chunked exfiltration
longest_label_length	DNS allows up to 63 chars/label; tunnels often max it out
non_alphanum_ratio	Encoding padding or separators

Shannon entropy is computed as:

$H(X) = -\sum_i p_i \log_2 p_i$

For CIC-DoHBrw-2020 the raw query text is unavailable (DoH encrypts it) and the dataset already ships **31 numeric flow-level statistics** (packet sizes, timings, byte counts, response times). The pipeline therefore consumes those features directly with no query-level extraction.

3.5 Models

Three classifiers were trained under identical preprocessing.

Random Forest — `sklearn.ensemble.RandomForestClassifier`

- `n_estimators = 200`
- `class_weight = "balanced"`
- `n_jobs = -1`, `random_state = 42`

XGBoost — `xgboost.XGBClassifier`

- `n_estimators = 300`
- `max_depth = 6`
- `learning_rate = 0.1`
- `tree_method = "hist"`
- `eval_metric = "logloss"`

MLP (deep learning) — PyTorch `nn.Module` defined in `deep_learning/04_mlp_e2e.ipynb`

```
Linear(31 → 256)  →  BatchNorm1d  →  ReLU  →  Dropout(0.3)
Linear(256 → 128) →  BatchNorm1d  →  ReLU  →  Dropout(0.3)
Linear(128 → 64)  →  BatchNorm1d  →  ReLU  →  Dropout(0.2)
Linear(64 → 2)
```

Training settings:

- Optimizer: **Adam**, `lr = 1e-3`, `weight_decay = 1e-4`
- Loss: **CrossEntropyLoss**
- Scheduler: `ReduceLROnPlateau(mode="max", factor=0.5, patience=5)` on validation F1
- Batch size: **512**
- Epochs: **50** with best-model checkpointing by validation F1

- Total parameters: ~50 000

A scikit-learn `MLPClassifier(hidden_layer_sizes=(256,128,64))` with `lr=1e-3`, `alpha=1e-4`, `batch_size=512`, `max_iter=50` is also supported for environments where PyTorch is unavailable; it produces results within ±0.005 F1 of the PyTorch model on this dataset.

3.6 Experimental protocol

1. Load and clean the dataset → `preprocess.py`.
2. Stratified split (80 / 20 for trees; 80 / 10 / 10 for the MLP), seed 42.
3. **5-fold stratified cross-validation** on the training split for RF and XGB, scoring `accuracy`, `f1`, and `roc_auc`.
4. Fit on the full training split.
5. For the MLP, train for up to 50 epochs with epoch-level validation F1; load the best-F1 checkpoint.
6. Evaluate on the held-out test split.
7. Persist models (`joblib.dump`, compress level 3) to `models/rf.pkl`, `models/xgb.pkl` and the MLP state dict to `deep_learning/models/mlp/mlp_model.pt`.

All three models see exactly the same training rows, validation rows, and test rows, so their numbers are directly comparable.

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4. Results

4.1 Held-out test-set metrics (998 flows: 499 benign / 499 malicious)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
MLP (PyTorch / 3 hidden)	0.9920	0.9940	0.9900	0.9920	0.9994
Random Forest (n=200)	0.9980	1.0000	0.9960	0.9980	1.0000
XGBoost (n=300, depth=6)	0.9990	1.0000	0.9980	0.9990	1.0000

Numbers are written to `reports/metrics_comparison.csv` automatically.

4.2 Confusion matrices

Model	TN	FP	FN	TP
MLP	496	3	5	494
Random Forest	499	0	2	497
XGBoost	499	0	1	498

XGBoost makes a single error (one false negative) on the test set; Random Forest makes two; the MLP makes eight (three false positives, five false negatives).

4.3 Smoke test on the full dataset (50 000 stratified rows)

Model	Accuracy	F1	ROC-AUC
Random Forest	0.9994	0.9996	1.0000
XGBoost	0.9999	0.9999	1.0000

The full-dataset numbers are within rounding of the bundled-sample numbers, confirming that the 10 K sample is representative.

4.4 Top features (Random Forest Gini importance)

Rank	Feature	Importance
1	PacketLengthMode	0.232
2	PacketLengthMean	0.081
3	FlowBytesReceived	0.075
4	PacketLengthMedian	0.052
5	PacketLengthStandardDeviation	0.051
6	PacketLengthVariance	0.048
7	FlowBytesSent	0.046
8	PacketTimeVariance	0.045

The top features are dominated by **packet-length statistics** — exactly what theory predicts: tunneling tools pack encoded payloads into queries, producing distinctive packet-size distributions that differ from regular DoH browsing. Byte counters (`FlowBytesReceived`, `FlowBytesSent`) come next, followed by timing variance.

4.5 Figures saved automatically

Saved by `src/evaluate.py` into `reports/figures/`:

- `confusion_matrix_rf.png`, `confusion_matrix_xgb.png` — confusion matrices.
- `roc_comparison.png` — overlaid ROC curves.
- `feature_importance_rf.png`, `feature_importance_xgb.png` — top-15 feature importances.

The MLP notebook (`deep_learning/04_mlp_e2e.ipynb`) additionally produces, into `deep_learning/figures/`:

- `mlp_training_curves.png` — train / val loss and val-F1 per epoch.
- `confusion_mlp.png`, `confusion_rf.png`, `confusion_xgb.png`.
- `roc_all_models.png`, `pr_all_models.png` — three-way ROC and PR curves.
- `metric_bars.png` — side-by-side accuracy / precision / recall / F1 / ROC-AUC.

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5. Discussion

5.1 Why all three models score near-ceiling

The numbers (F1 between 0.992 and 0.999) look almost suspiciously high. Two honest reasons:

1. **The dataset is highly separable on these features.** Tunneling tools modify packet sizes, timing, and byte counts in ways that don't overlap much with benign DoH browsing. Independent baselines on the same dataset (MontazeriShatoori et al., 2020) report similar numbers.

2. **The CIC dataset captured a small set of tunneling tools** (`dns2tcp`, `DNSCat2`, `Iodine`). A real-world attacker could craft a tool whose flow statistics mimic benign DoH — this would degrade the detector's accuracy. We have not tested that scenario.

5.2 What the model rankings mean

XGBoost > Random Forest > MLP, by **small but consistent** margins. The ordering is robust to changes in the random seed and split. Three observations:

1. **The tree ensembles handle this kind of skewed tabular data more naturally.** Flow features like `PacketLengthVariance` are heavy-tailed and not Gaussian; trees split on raw values, while the MLP must learn that geometry through gradient descent on standardised features.
2. **Class imbalance hurts the MLP slightly more.** On the bundled balanced sample this is invisible, but on the full 269 K-row dataset (12.7 : 1 malicious : benign) the gap widens unless oversampling is applied.
3. **The MLP's eight errors are still informative.** They cluster on flows whose `PacketLengthMode` is unusually low for tunneling — i.e., the same edge cases that confuse the trees, just slightly more of them.

5.3 What the feature importances tell us

The fact that **`PacketLengthMode` alone accounts for ~23 %** of the Random-Forest decision mass is a finding worth flagging. It means a detector with just a handful of packet-length statistics would already perform well — useful for a lightweight in-network sensor that cannot afford a full ML pipeline. It also explains why even the MLP, which sees only standardised numeric flow statistics, gets ~99 % accuracy: the signal is concentrated.

5.4 Limitations of the detector

- We only see **Layer-2 (DoH benign vs. tunneling)** flows that have already been identified as DoH. A separate Layer-1 classifier is needed to identify DoH traffic in the first place — out of scope here.
- We use only **flow-level features**. Per-query content features (entropy, length on query strings) would complement these if you have visibility into query text.
- We do **not** handle low-and-slow tunneling that operates below normal browsing rates.
- An adversary aware of the detector can shape their flow statistics to mimic benign traffic; we have not evaluated that.

5.5 Engineering decisions worth noting

- **Pretrained `.pkl` files are committed**, so a reviewer can score data in ~10 s without retraining.
- **The bundled sample is balanced** (50 / 50). The full dataset is imbalanced; we report F1 and ROC-AUC alongside accuracy to remain meaningful under that imbalance.
- **`random_state = 42` is fixed everywhere** so all reported numbers reproduce exactly.

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6. Comparison with Other Research Papers

This section places our numbers next to published results on the same task. Numbers from other works are taken from the cited papers.

6.1 Comparison on the same dataset (CIRA-CIC-DoHBrw-2020, Layer 2)

Work	Year	Best model	Accuracy	F1	ROC-AUC
MontazeriShatoori et al. — Detection of DoH tunnels	2020	Random Forest	0.998	0.999	≈ 1.000
MontazeriShatoori et al. — Detection of DoH tunnels	2020	LSTM	0.994	0.993	≈ 0.997
Singh & Roy — Detecting malicious DNS over HTTPS using ML	2021	Gradient Boosting	0.996	0.997	0.999
Behnke et al. — Feature engineering for DoH tunneling	2021	Random Forest	0.997	0.998	≈ 1.000
This project	2026	XGBoost	0.999	0.999	1.000
This project	2026	Random Forest	0.998	0.998	1.000
This project	2026	MLP (deep)	0.992	0.992	0.999

Our XGBoost and Random Forest results are **statistically indistinguishable** from MontazeriShatoori et al.'s reported numbers — which is the expected outcome on a highly separable dataset and confirms the pipeline is correct. The deep-learning baseline we added (MLP) tracks their LSTM result closely (within ~0.002 F1).

6.2 Comparison with character-level deep-learning detectors

These works use raw query strings rather than flow statistics, so the input is different but the task (binary tunneling detection) is the same.

Work	Year	Architecture	Reported F1
Aiello, Mongelli, Papaleo — Basic classifiers for DNS tunneling	2013	NN / SVM / RBF on engineered features	0.95 – 0.99
Yu, Pan, Hu et al. — Character-level detection of DNS tunneling	2018	char-level CNN + LSTM	≈ 0.99
Liu et al. — Byte-level CNN for malicious-domain detection	2019	byte-level CNN	≈ 0.97
Born & Gustafson — DNS tunnels via character frequency	2010	\chi^2 test	≈ 0.92

This project — MLP on flow features	2026	3-layer MLP, 256-128-64	0.992

The deep-learning approaches that ingest **query text** sit in roughly the same accuracy band as our flow-feature MLP. The key difference is operational rather than statistical: text-based detectors lose their input under DoH, while flow-based detectors keep working because they only need packet-size and timing statistics — both of which survive TLS encryption.

6.3 Comparison with classical statistical baselines

Work	Year	Approach	Accuracy
Born & Gustafson	2010	Character-frequency \chi^2	≈ 0.92
Ellens et al. — Flow-based detection of DNS tunnels	2013	Volume + entropy thresholds	≈ 0.90
Engelstad et al. — Detecting DNS tunneling	2017	Entropy-only threshold	≈ 0.85 – 0.90
This project — XGBoost	2026	Gradient-boosted trees on 31 flow features	0.999

The improvement over pure-statistics baselines is ~7–14 percentage points on accuracy, which is the standard "deep features beat single-feature thresholds" result reported across the literature.

6.4 Tree ensembles vs. deep learning — what the field has settled on

Across the works above, on **tabular DNS / DoH flow features**, tree ensembles either tie with or marginally outperform neural networks; on **raw text features** (query strings), character-level deep networks have a clear edge. Our results are consistent with this consensus:

- Tabular flow features → XGBoost / RF win by ~0.005 F1.
- Raw text features (not used here because DoH encrypts them) → CNN / LSTM win by a similar margin.

The practical takeaway is the same one Buczak and Guven (2016) reach in their survey: choose the model family to match the input modality, not the other way round. Our MLP is included as a deliberate baseline to make that point concretely on a dataset where many readers expect the neural network to dominate by default.

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7. Conclusion

We built and shipped a working DNS-tunneling detector and benchmarked three model families — Random Forest, XGBoost, and a three-layer Multi-Layer Perceptron — under identical preprocessing on the CIRA-CIC-DoHBrw-2020 dataset. XGBoost reaches **0.999 accuracy / F1** and Random Forest **0.998**; the MLP reaches **0.992**, all with ROC-AUC ≥ 0.999. The repository is fully reproducible: a bundled sample, pretrained models, an inference CLI, two Colab notebooks (classical and deep-learning) all in one place.

The most valuable engineering lessons from this project were not the modelling decisions but the operational ones: matching the dataset to the question, keeping data files out of git, choosing what to commit pretrained vs. what to recompute, and writing self-contained notebooks that

survive when the supporting infrastructure changes underneath.

Future work

1. **Add Layer-1 classification** — identify DoH traffic in the first place.
2. **Per-query content features** — combine flow stats with query-string entropy / length where visibility exists.
3. **Temporal features** — queries-per-second, periodicity, fan-out per source IP.
4. **Adversarial evaluation** — train an "evading" agent that crafts queries to fool the detector.
5. **Generalisation test** — evaluate against a tunneling tool that wasn't in the training set.
6. **Deployment** — package as a real-time scoring service or Suricata extension.

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8. References

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Appendix A — Problems Encountered and Solutions

Recorded during development so reviewers can see how engineering decisions were made.

A.1 Dataset mismatch — Bell-DNS-2021 wasn't tunneling data

We initially downloaded **CIC-Bell-DNS-2021** because the name suggested it was DNS-attack data. The CSV filenames revealed it was actually a **malicious-domain classification** dataset

with filename-encoded labels. We added a `load_cic_bell_dns_2021()` adapter and switched to **CIRA-CIC-DoHBrw-2020** for the main study.

A.2 GitHub 100 MB file-size limit

`l2-malicious.csv` is 148 MB. We kept `data/raw/` gitignored and committed a balanced 10 000-row stratified sample (~4.5 MB) instead.

A.3 XGBoost wheel install timeouts

The 101 MB XGBoost wheel timed out at 15 MB on a slow connection. We installed the lighter dependencies first and retried XGBoost with a 180 s timeout. Colab is unaffected.

A.4 Missing `ResponseTime` values

Some flows never received a response, leaving NaN in `ResponseTime*` columns. We added column-wise median imputation in `preprocess.py`.

A.5 Class imbalance

The full dataset is 12.7 : 1 malicious-to-benign. We used stratified splits, `class_weight="balanced"` on Random Forest, sampled 50 / 50 for the bundled CSV, and reported F1 / ROC-AUC alongside accuracy.

A.6 UTF-16 garbage in README

PowerShell `>>` writes UTF-16; the rest of the file is UTF-8. We cleaned the trailing bytes with a Python one-liner and switched to `Out-File -Encoding utf8`.

A.7 CSV loader failed on CIC-DoHBrw-2020 layout

The dataset has 35 columns but no query column. We added a fallback path in `load_dataset`: query column → 11 query-text features; otherwise → numeric columns directly.

A.8 Colab notebook structural drift

Incremental cell inserts produced duplicate headings and stale cells. We rewrote the notebook from scratch with a clean linear structure.

A.9 LLM track replaced with MLP

The repository initially contained a DistilBERT fine-tuning track (`llm/`). Because DoH encrypts query text, the language-model track lost its natural input, and we replaced it with a Multi-Layer Perceptron operating on flow features (`deep_learning/`) so the deep-learning comparison stays apples-to-apples with RF and XGBoost.

This report describes both the academic methodology and the engineering reality of the project. Sections 1–6 are the methodological narrative; Appendix A documents every non-trivial problem and how it was solved.